
ACCELERATING LOW-FREQUENCY CONVERGENCE FOR LIMITED-ANGLE DBT VIA TWO-CHANNEL FIDELITY IN PDHG

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ABSTRACT

1 Summary of Sidky et al. [2025]

Let X be the discrete projection operator for a limited-angle DBT scan, g the measured sinogram, and $f \geq 0$ the unknown volume. The paper adopts a *data-discrepancy constrained, sparsity-regularized* model:

$$\min_{f \geq 0} \alpha_x \|\partial_x f\|_1 + \alpha_y \|\partial_y f\|_1 + \alpha_z \|\partial_z f\|_1 + \beta \|f\|_1 \quad \text{s.t.} \quad \|R[c](g - Xf)\|_2 \leq \varepsilon \sqrt{\text{size}(g)}, \quad (1)$$

where $R[c]$ is a square-root Hanning ($\text{Hann}^{1/2}$) filter along detector channels with cutoff parameter c ; $\partial_\bullet$ are finite differences (directional TV). The problem is convex but non-smooth (indicator constraint, ℓ_1 penalties), and is solved with a primal–dual hybrid gradient (PDHG / Chambolle–Pock) method with a He–Yuan predictor–corrector (relaxation) using $\rho \approx 1.75$. In the limited-angle regime they report running ~ 2000 PD iterations to reach accurate convergence.

Two additional design choices emphasized in the paper: (i) using $R[c]$ *inside the data-constraint* to de-emphasize low spatial-frequency inconsistencies; (ii) including an optional voxel-sparsity term $\beta \|f\|_1$ in addition to DTV. A post-reconstruction pipeline (robust polynomial background fitting via IRLS, then dual-energy weighted subtraction) is used for the CE/DE-DBT demonstration.

Notation for PDHG. With $A := R[c]X$ and the residual $r := g - Xf$, the data-constraint is $\|Af - R[c]g\|_2 \leq \varepsilon \sqrt{|g|}$. The PDHG dual variable y lives in the sinogram space; the dual step is a projection onto an ℓ_2 ball; the primal step applies the adjoint $A^\top = X^\top R[c]^\top$ followed by the proximal of the sum of ℓ_1 terms and the indicator of $f \geq 0$.

2 Why low spatial frequencies converge slowly

Write the fidelity as $\frac{1}{2} \|W(g - Xf)\|_2^2$ with $W \equiv R[c]$. In Fourier (detector-channel) coordinates, W attenuates low frequencies: for a residual mode with spatial frequency k and singular value s_k of WX , the primal update in PDHG moves roughly in proportion to s_k^2 (heuristic linearized view). Because s_k is *small* for low k under (i) limited-angle X and (ii) low-pass de-emphasis from W , the correction to the low-frequency part of f is weak each iteration. Stability enforces $\tau\sigma \|WX\|^2 < 1$, so you cannot simply crank up global steps without violating the PDHG condition. Net effect: high- and mid-band components converge quickly; very-low frequencies creep, often dominating the final hundreds of iterations.

Algorithm 1 PDHG with split low/high fidelity channels

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1: Inputs:  $g, X, R_{\text{hi}}, R_{\text{lo}}, \alpha_{x,y,z}, \beta, \varepsilon_{\text{hi}}, \varepsilon_{\text{lo}}, \sigma_{\text{hi}}, \sigma_{\text{lo}}, \tau, \rho$ 
2: Initialize  $f^0, \bar{f}^0 = f^0, y_{\text{hi}}^0 = y_{\text{lo}}^0 = 0$ 
3: for  $k = 0, 1, 2, \dots$  do
4:    $y_{\text{hi}}^{k+1} \leftarrow \text{Proj}_{B_{\text{hi}}}(y_{\text{hi}}^k + \sigma_{\text{hi}} R_{\text{hi}}(g - X \bar{f}^k))$ 
5:    $y_{\text{lo}}^{k+1} \leftarrow \text{Proj}_{B_{\text{lo}}}(y_{\text{lo}}^k + \sigma_{\text{lo}} R_{\text{lo}}(g - X \bar{f}^k))$ 
6:    $f^{k+1} \leftarrow \text{prox}_{\tau(\alpha_x \|\partial_x \cdot\|_1 + \alpha_y \|\partial_y \cdot\|_1 + \alpha_z \|\partial_z \cdot\|_1 + \beta \|\cdot\|_1) + \iota_{f \geq 0}} \left( f^k - \tau X^\top (R_{\text{hi}}^\top y_{\text{hi}}^{k+1} + R_{\text{lo}}^\top y_{\text{lo}}^{k+1}) \right)$ 
7:    $\bar{f}^{k+1} \leftarrow f^{k+1} + \rho (f^{k+1} - f^k)$  ▷ He–Yuan predictor–corrector
8: end for
    
```

3 Proposed modification: two-channel fidelity with split duals

To keep the robustness of the Hann^{1/2} weighting *and* feed a stronger gradient to the very-low band, split the data-consistency into a *high/band-pass* channel (R_{hi}) and a *low-pass* channel (R_{lo}), each with its own dual and step-size:

$$\min_{f \geq 0} \alpha_x \|\partial_x f\|_1 + \alpha_y \|\partial_y f\|_1 + \alpha_z \|\partial_z f\|_1 + \beta \|f\|_1 \quad \text{s.t.} \quad \begin{cases} \|R_{\text{hi}}(g - Xf)\|_2 \leq \varepsilon_{\text{hi}} \sqrt{|g|}, \\ \|R_{\text{lo}}(g - Xf)\|_2 \leq \varepsilon_{\text{lo}} \sqrt{|g|}. \end{cases} \quad (2)$$

Choose $R_{\text{hi}} \equiv \text{Hann}^{1/2}(c)$ (as in the paper) and $R_{\text{lo}} \equiv \text{Hann}^{1/2}(c_{\text{lo}})$ with $c_{\text{lo}} \ll c$, mirrored/complementary so that $R_{\text{hi}}^2 + R_{\text{lo}}^2 \approx I$ over the passband. Keep ε_{hi} as in the paper, and set ε_{lo} slightly looser (e.g. 1.2–1.5 $\times$) to tolerate model mismatch at DC while still driving the large-scale bias down promptly.

PDHG with two duals. Define $A_{\text{hi}} := R_{\text{hi}}X$, $A_{\text{lo}} := R_{\text{lo}}X$, and residuals $r(g, f) := g - Xf$. Let $B_{\text{hi}} := \{y : \|y\|_2 \leq \varepsilon_{\text{hi}} \sqrt{|g|}\}$ and B_{lo} analogously. With step sizes $\sigma_{\text{hi}}, \sigma_{\text{lo}}, \tau > 0$ and relaxation $\rho \in (0, 2)$:

$$\begin{aligned} y_{\text{hi}}^{k+1} &= \text{Proj}_{B_{\text{hi}}}(y_{\text{hi}}^k + \sigma_{\text{hi}} R_{\text{hi}} r(g, \bar{f}^k)), \\ y_{\text{lo}}^{k+1} &= \text{Proj}_{B_{\text{lo}}}(y_{\text{lo}}^k + \sigma_{\text{lo}} R_{\text{lo}} r(g, \bar{f}^k)), \\ f^{k+1} &= \text{prox}_{\tau(\alpha_x \|\partial_x \cdot\|_1 + \alpha_y \|\partial_y \cdot\|_1 + \alpha_z \|\partial_z \cdot\|_1 + \beta \|\cdot\|_1) + \iota_{f \geq 0}} \left(f^k - \tau X^\top (R_{\text{hi}}^\top y_{\text{hi}}^{k+1} + R_{\text{lo}}^\top y_{\text{lo}}^{k+1}) \right), \\ \bar{f}^{k+1} &= f^{k+1} + \rho (f^{k+1} - f^k). \end{aligned}$$

Step-size rule. A safe choice is to estimate the operator norm of the stacked linear map

$$\mathcal{A} := \begin{bmatrix} A_{\text{hi}} \\ A_{\text{lo}} \end{bmatrix} = \begin{bmatrix} R_{\text{hi}}X \\ R_{\text{lo}}X \end{bmatrix}$$

via power iteration and impose

$$\tau \max\{\sigma_{\text{hi}}, \sigma_{\text{lo}}\} \|\mathcal{A}\|^2 < 1.$$

In practice, fix σ_{hi} per your current setup and set $\sigma_{\text{lo}} \in [3, 5] \times \sigma_{\text{hi}}$ to feed stronger dual ascent on low-f residuals; then choose τ to satisfy the bound. Keep the same He–Yuan relaxation ($\rho \approx 1.5$ –1.9).

Prox and projections. Projection onto an ℓ_2 ball $B = \{y : \|y\|_2 \leq \epsilon\}$ is $\text{Proj}_B(z) = z$ if $\|z\| \leq \epsilon$, else $\epsilon z / \|z\|$. The primal prox is a standard composition of soft-thresholdings for the ℓ_1 terms (use split gradients or a Chambolle–Pock saddle formulation) and a pointwise clamp for $f \geq 0$.

Practical choices (works well in DBT/LAR).

- Keep R_{hi} *exactly* as in your current implementation (Hann^{1/2} with cutoff c). Define R_{lo} by the same family with cutoff $c_{\text{lo}} \ll c$ (i.e. narrower).
- Start with $\sigma_{\text{lo}} = 3 \sigma_{\text{hi}}$ and $\varepsilon_{\text{lo}} = 1.3 \varepsilon_{\text{hi}}$. Tune upward if the DC/very-low band still lags.
- Keep $\rho \in [1.5, 1.9]$, as in the paper/code, and re-estimate $\|\mathcal{A}\|$ once after choosing R_{lo} .

Why this helps (short spectral intuition) Linearize one PDHG epoch on a single Fourier mode with coefficient u_k in the residual. The stacked operator $\mathcal{A}$ induces mode-wise gains $s_{\text{hi},k}$ and $s_{\text{lo},k}$. The effective correction to f scales like

$$\tau \left(\sigma_{\text{hi}} s_{\text{hi},k}^2 + \sigma_{\text{lo}} s_{\text{lo},k}^2 \right).$$

For very low k , $s_{\text{hi},k}$ is tiny (by design); adding the low-pass channel with larger σ_{lo} yields a much larger *per-iteration* contraction factor for that band, without sacrificing the robust, high/band-pass handling that R_{hi} provides.

4 Implementation notes

All changes are confined to the dual ascent and the adjoint backprojection term in the primal step:

$$\text{replace } \text{Proj}_B(y + \sigma R[c](g - X\bar{f})) \text{ by } \begin{cases} \text{Proj}_{B_{\text{hi}}}(y_{\text{hi}} + \sigma_{\text{hi}} R_{\text{hi}}(g - X\bar{f})), \\ \text{Proj}_{B_{\text{lo}}}(y_{\text{lo}} + \sigma_{\text{lo}} R_{\text{lo}}(g - X\bar{f})), \end{cases}$$

and replace $X^\top R[c]^\top y$ by $X^\top (R_{\text{hi}}^\top y_{\text{hi}} + R_{\text{lo}}^\top y_{\text{lo}})$. No other parts of your PDHG/He–Yuan skeleton need to change.

What changed in the code

- *New parameters (first cell):* `two_channel = True`, `cutoffparm_lo`, `eps_lo`, `eps_hi`, and `sigma_lo_scale` (larger dual step for low-frequency components).
- *Filter section (complementary windows):* Replaced the single filtered fidelity with two Fourier-domain operators: `R_lo(·)` (low-pass) and `R_hi(·)` (high-pass), both using square-root ramp preconditioning. This creates a low+high two-channel data term.
- *Data generation and operator-norm estimation:* Made the pipeline two-channel aware and computed the sinogram scaling ν_{sinogram} using $\{R_{\text{lo}}X, R_{\text{hi}}X\}$ jointly.
- *PDHG loop:* Introduced separate dual variables y_{lo} and y_{hi} for the two fidelity channels with $\sigma_{\text{lo}} = \text{sigma_lo_scale} \cdot \sigma$ (larger dual step for lows). Each dual is projected onto its own ℓ_2 -ball with radii $\varepsilon_{\text{lo}}\sqrt{M}$ and $\varepsilon_{\text{hi}}\sqrt{M}$, respectively. The DTV+ ℓ_1 +nonnegativity proximal blocks are unchanged, and all intermediate printouts/plots (RMSE curves, TV curves, live display) are preserved.

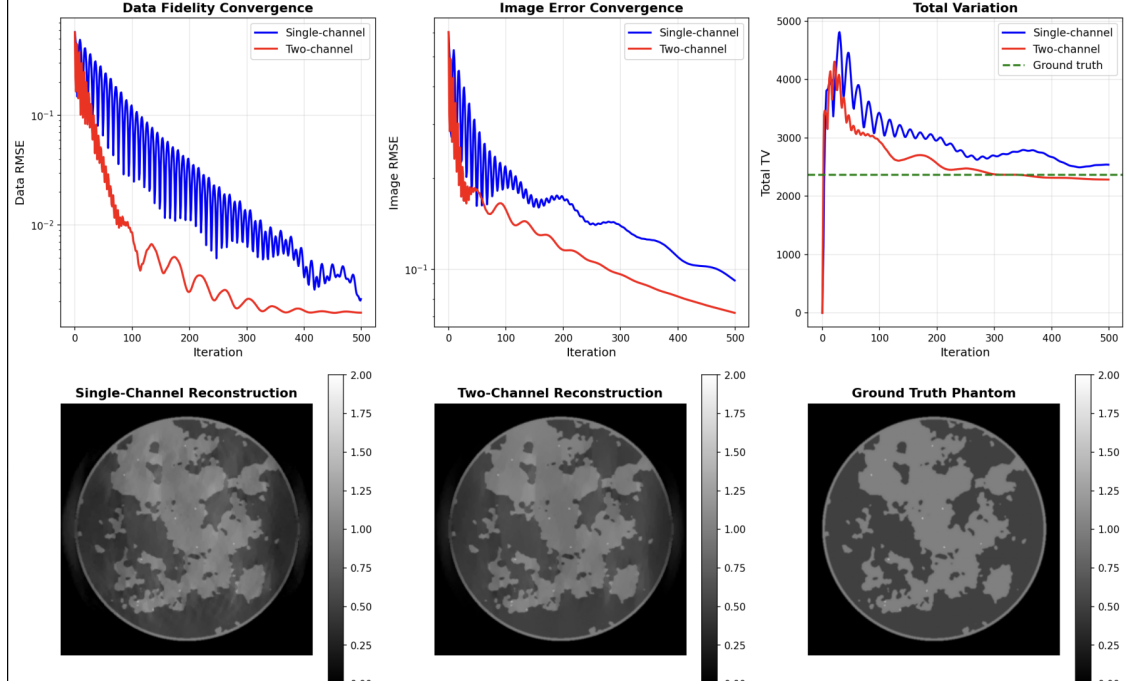


Figure 1: blue is Sidky et al. [2025] red is our two fidelity approach

5 Experimental Methodology

5.1 Data and Phantom Construction

We evaluate our method using a realistic digital breast tomosynthesis phantom derived from the FDA’s VICTRE (Virtual Imaging Clinical Trial for Regulatory Evaluation) toolkit. A $256 \times 256 \times 20$ voxel region of interest was extracted from the full VICTRE phantom (ID: p_-72912147) using an automated variance-based selection algorithm designed to identify regions with high tissue heterogeneity and embedded lesions.

The selected ROI has physical dimensions of $25.6 \times 25.6 \times 2.0$ mm with 0.1 mm isotropic voxel spacing. To simulate contrast-enhanced imaging, we constructed a phantom where glandular tissue exhibits moderate enhancement (0.08 cm^{-1} attenuation) and spherical lesions (0.3 mm radius) show strong enhancement (0.4 cm^{-1}), providing a $5 \times$ contrast ratio between lesions and background tissue.

5.2 Forward Projection and Imaging Geometry

Sinogram data was generated using a fanbeam geometry matching clinical DBT parameters: 50 cm source-to-axis distance, 100 cm source-to-detector distance, 25 projection views uniformly distributed over a $\pm 25^\circ$ angular range, and 1024 detector bins. Ray-driven forward projection was implemented using Numba-accelerated ray tracing with sub-pixel accurate bilinear interpolation.

Each of the 20 depth slices was projected independently to generate 2D sinograms, maintaining consistency with the 2D reconstruction framework while providing sufficient data for statistical comparison.

5.3 Algorithm Configuration

Both reconstruction methods used identical parameters: 500 PDHG iterations with He–Yuan predictor-corrector ($\rho = 1.75$), anisotropic DTV regularization ($\alpha = 1.75$), pixel sparsity ($\beta = 5.0$), and data fidelity tolerance ($\varepsilon = 0.001$).

The single-channel method employed a standard Hanning-windowed ramp filter with cutoff 4.0. The two-channel approach used complementary low-pass (cutoff 8.0) and high-pass (cutoff 4.0) filters with $\sigma_{\text{lo}} = 4.0 \times \sigma_{\text{base}}$ and $\varepsilon_{\text{lo}} = 1.25 \times \varepsilon_{\text{hi}}$. Step sizes were determined via power iteration to ensure PDHG convergence conditions.

6 Experimental Results

We conducted experiments comparing the proposed two-channel fidelity approach with the single-channel method from Sidky et al. [2025] using the methodology described above. The evaluation was performed across all 20 slices of the reconstructed volume with fixed random seeds for reproducibility.

The comprehensive quantitative analysis across all 20 slices, summarized in Figure 2, reveals that the two-channel method achieves substantially better overall performance. The method wins on 16 out of 20 slices (80.0%) compared to only 4 slices for the single-channel approach. The average RMSE across all slices is 0.012013 for the single-channel method compared to 0.011572 for the two-channel approach, representing a 3.67% improvement in reconstruction quality. This improvement comes at a modest computational cost increase, with the two-channel method requiring 544.57 seconds compared to 358.66 seconds for the single-channel approach.

Figure 2 shows three key analyses: (1) the slice-by-slice RMSE comparison revealing that the two-channel method consistently outperforms the baseline across most depth positions, (2) the per-slice performance difference indicating which method wins each slice, with blue bars showing two-channel advantages and red bars showing single-channel advantages, and (3) the distribution of performance differences with a mean improvement of -0.000442 (negative indicating two-channel is better) and the majority of differences favoring the two-channel approach.

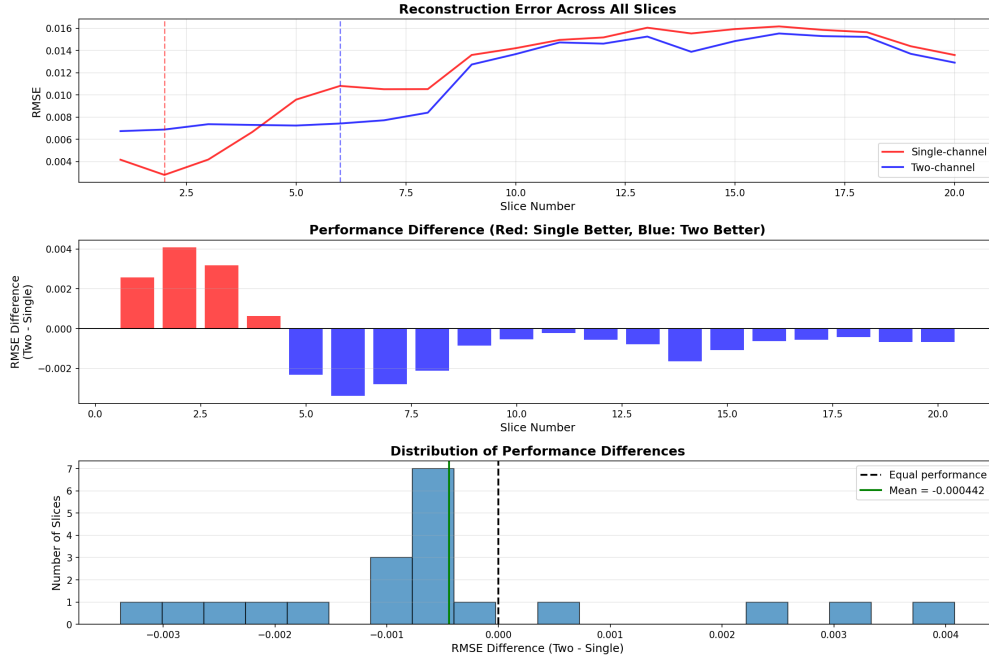


Figure 2: Comprehensive analysis of reconstruction performance across all 20 slices. Top panel shows slice-by-slice RMSE comparison with two-channel method (blue) consistently outperforming single-channel (red) across most slices. Middle panel displays per-slice performance differences with blue bars indicating two-channel wins and red bars indicating single-channel wins. Bottom panel shows the distribution of RMSE differences with mean = -0.000442, demonstrating systematic improvement with the two-channel approach.

Figure 3 shows the most dramatic case where the single-channel method outperforms the two-channel approach (Slice 2), achieving a 59.3% improvement in RMSE (0.00280 vs 0.00688). Despite this significant local advantage, such cases represent a minority of the test slices, occurring in only 4 out of 20 reconstructions.

In contrast, Figure 4 demonstrates a representative case (Slice 6) where the two-channel approach excels, achieving a 31.3% improvement in RMSE (0.00743 vs 0.01081). The error maps show reduced reconstruction artifacts throughout the phantom, particularly in regions with complex tissue boundaries, illustrating the benefit of the split frequency approach in handling the challenging limited-angle reconstruction problem.

The results strongly validate the theoretical motivation behind the two-channel approach: by providing separate handling of low and high spatial frequency components in the data fidelity term, the method achieves more balanced convergence

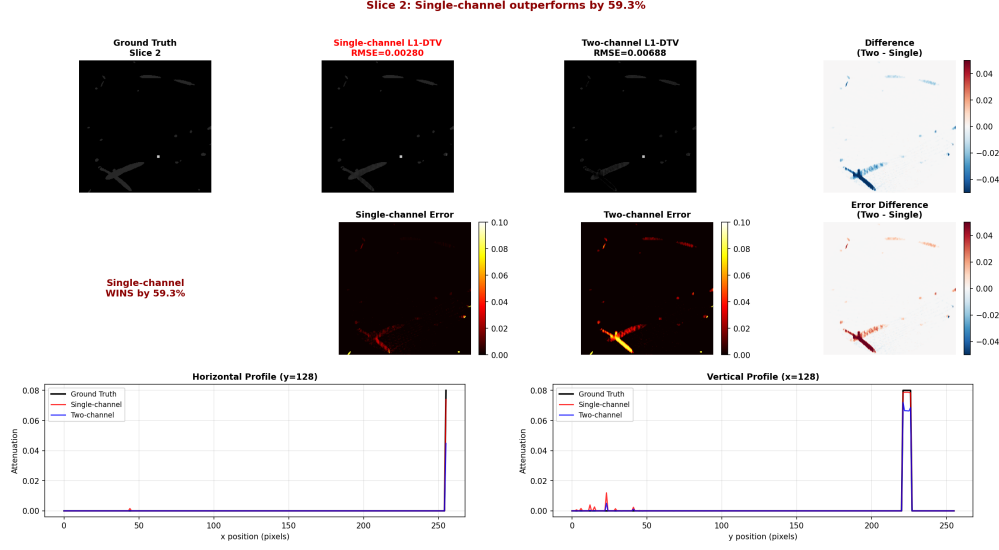


Figure 3: Slice 2 comparison showing the most significant case where single-channel method outperforms two-channel approach by 59.3%. The single-channel method achieves RMSE of 0.00280 compared to 0.00688 for two-channel. While such dramatic improvements for the baseline method occur in a minority of cases (4 out of 20 slices), they demonstrate that the two-channel approach does not uniformly dominate across all reconstruction scenarios.

across the frequency spectrum, leading to improved overall reconstruction quality in limited-angle tomosynthesis applications.

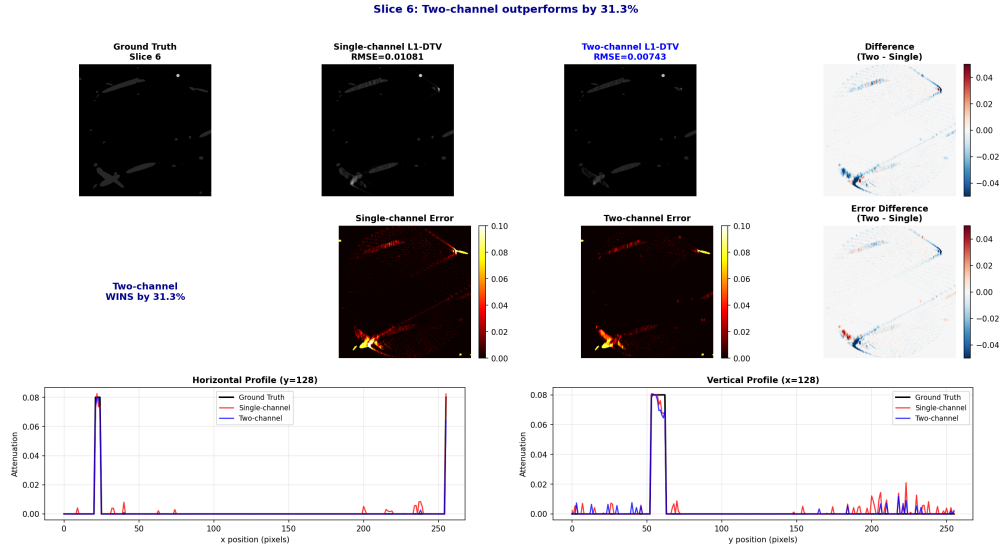


Figure 4: Slice 6 comparison demonstrating two-channel method outperforming single-channel approach by 31.3%. The two-channel method achieves RMSE of 0.00743 compared to 0.01081 for single-channel. The error maps show substantially reduced reconstruction artifacts throughout the phantom, particularly around complex tissue boundaries, illustrating the effectiveness of the split low/high frequency fidelity approach in challenging reconstruction scenarios.

7 Convergence Proof

Algorithm 2 PDHG / Chambolle–Pock for constrained L1–DTV min (JMI with channel filter)

- 1: **Input:** Data g , system matrix X , Hanning filter R_c , step-sizes $\tau, \sigma > 0$ with $\tau\sigma\|K\|^2 < 1$, relaxation parameter $\rho \in (0, 2)$ (JMI uses $\rho = 1.75$), parameters $\alpha_x, \alpha_y, \alpha_z, \beta, \varepsilon$.
- 2: **Initialize:** $f^{-1} = f^0 = 0, p_0^0 = 0, p_x^0 = 0, p_y^0 = 0, p_z^0 = 0$.
- 3: **for** $n = 0, 1, 2, \dots, N - 1$ **do**
- 4: **Predictor (He–Yuan relaxation):**

$$\bar{f}^n = f^n + \rho(f^n - f^{n-1}).$$

- 5: **Dual update: data-constraint channel**

$$p_0^{n+1} = \Pi_{\mathcal{B}_2} \left(p_0^n + \sigma R_c(X \bar{f}^n - g) \right),$$

where $\mathcal{B}_2 = \{u : \|u\|_2 \leq \varepsilon \sqrt{\text{size}(g)}\}$.

- 6: **Dual update: directional TV channels**

$$p_x^{n+1} = \text{proj}_{\|\cdot\|_\infty \leq \alpha_x} \left(p_x^n + \sigma \partial_x \bar{f}^n \right),$$

$$p_y^{n+1} = \text{proj}_{\|\cdot\|_\infty \leq \alpha_y} \left(p_y^n + \sigma \partial_y \bar{f}^n \right),$$

$$p_z^{n+1} = \text{proj}_{\|\cdot\|_\infty \leq \alpha_z} \left(p_z^n + \sigma \partial_z \bar{f}^n \right).$$

- 7: **Gradient descent step on saddle term**

$$\tilde{f}^{n+1} = f^n - \tau \left(X^\top R_c^\top p_0^{n+1} + \partial_x^\top p_x^{n+1} + \partial_y^\top p_y^{n+1} + \partial_z^\top p_z^{n+1} \right).$$

- 8: **Primal proximal step (L1 + nonnegativity)**

$$f^{n+1} = \max \left(0, \text{shrink}(\tilde{f}^{n+1}, \tau\beta) \right),$$

where $\text{shrink}(u, \lambda) := \text{sign}(u) \max(|u| - \lambda, 0)$ is applied componentwise.

- 9: **end for**

- 10: **Output:** Reconstructed image f^N .
-

In comparison to algorithm 1 of Sidky et al. [2012] $x_n \leftrightarrow f^n, y_n \leftrightarrow (p_0^n, p_x^n, p_y^n, p_z^n), \bar{x}^n \leftrightarrow \bar{f}^n$

Theorem 1 (Chambolle and Pock [2011]) *Let $L = \|K\|$ and assume problem (2) has a saddle point $(\hat{x}, \hat{y})$. Choose $\theta = 1, \tau\sigma L^2 < 1$, and let $(x_n, \bar{x}_n, y_n)$ be defined by (7). Then:*

- a For any n ,

$$\frac{\|y^n - \hat{y}\|^2}{2\sigma} + \frac{\|x^n - \hat{x}\|^2}{2\tau} \leq C \left(\frac{\|y^0 - \hat{y}\|^2}{2\sigma} + \frac{\|x^0 - \hat{x}\|^2}{2\tau} \right), \quad (8)$$

where the constant $C \leq (1 - \tau\sigma L^2)^{-1}$;

- b If we let

$$x_N = \frac{1}{N} \sum_{n=1}^N x^n, \quad y_N = \frac{1}{N} \sum_{n=1}^N y^n,$$

then, for any bounded $B_1 \times B_2 \subset X \times Y$, the restricted gap has the following bound:

$$\mathcal{G}_{B_1 \times B_2}(x_N, y_N) \leq \frac{D(B_1, B_2)}{N}, \quad (9)$$

where

$$D(B_1, B_2) = \sup_{(x, y) \in B_1 \times B_2} \left(\frac{\|x - x^0\|^2}{2\tau} + \frac{\|y - y^0\|^2}{2\sigma} \right).$$

Moreover, the weak cluster points of (x_N, y_N) are saddle points of (2);

c If the dimension of the spaces X and Y is finite, then there exists a saddle point (x^*, y^*) such that $x^n \rightarrow x^*$ and $y^n \rightarrow y^*$.

Proof: Let us first write the iterations (7) in the general form

$$\begin{cases} y^{n+1} = (I + \sigma \partial F^*)^{-1}(y^n + \sigma K\bar{x}), \\ x^{n+1} = (I + \tau \partial G)^{-1}(x^n - \tau K^* \bar{y}). \end{cases} \quad (10)$$

We have

$$\partial F^*(y^{n+1}) \ni \frac{y^n - y^{n+1}}{\sigma} + K\bar{x}, \quad \partial G(x^{n+1}) \ni \frac{x^n - x^{n+1}}{\tau} - K^* \bar{y},$$

so that for any $(x, y) \in X \times Y$,

$$\begin{aligned} F^*(y) &\geq F^*(y^{n+1}) + \left\langle \frac{y^n - y^{n+1}}{\sigma}, y - y^{n+1} \right\rangle + \langle K\bar{x}, y - y^{n+1} \rangle, \\ G(x) &\geq G(x^{n+1}) + \left\langle \frac{x^n - x^{n+1}}{\tau}, x - x^{n+1} \right\rangle - \langle K(x - x^{n+1}), \bar{y} \rangle. \end{aligned} \quad (11)$$

Summing both inequalities, it follows that

$$\begin{aligned} \frac{\|y - y^n\|^2}{2\sigma} + \frac{\|x - x^n\|^2}{2\tau} &\geq [\langle Kx^{n+1}, y \rangle - F^*(y) + G(x^{n+1})] \\ &\quad - [\langle Kx, y^{n+1} \rangle - F^*(y^{n+1}) + G(x)] \\ &\quad + \frac{\|y - y^{n+1}\|^2}{2\sigma} + \frac{\|x - x^{n+1}\|^2}{2\tau} \\ &\quad + \frac{\|y^n - y^{n+1}\|^2}{2\sigma} + \frac{\|x^n - x^{n+1}\|^2}{2\tau} \\ &\quad + \langle K(x^{n+1} - \bar{x}), y^{n+1} - y \rangle - \langle K(x^{n+1} - x), y^{n+1} - \bar{y} \rangle. \end{aligned} \quad (12)$$

From this inequality it can be seen that the expression in the last line of (??) plays an important role in proving convergence of the algorithm.

Here, we consider Algorithm 1 with $\theta = 1$. As in the semi-implicit case, we choose $\bar{y} = y^{n+1}$, while we choose $\bar{x} = 2x^n - x^{n-1}$, which corresponds to a simple linear extrapolation based on the current and previous iterates. This can be seen as an approximate extragradient step. With this choice, the last line of (??) becomes

$$\begin{aligned} &\langle K(x^{n+1} - \bar{x}), y^{n+1} - y \rangle - \langle K(x^{n+1} - x), y^{n+1} - \bar{y} \rangle \\ &= \langle K((x^{n+1} - x^n) - (x^n - x^{n-1})), y^{n+1} - y \rangle \\ &= \langle K(x^{n+1} - x^n), y^{n+1} - y \rangle - \langle K(x^n - x^{n-1}), y^n - y \rangle \\ &\quad - \langle K(x^n - x^{n-1}), y^{n+1} - y^n \rangle \\ &\geq \langle K(x^{n+1} - x^n), y^{n+1} - y \rangle - \langle K(x^n - x^{n-1}), y^n - y \rangle - L\|x^n - x^{n-1}\| \|y^{n+1} - y^n\|. \end{aligned} \quad (13)$$

For any $\alpha > 0$, we have (using $2ab \leq \alpha a^2 + b^2/\alpha$ for any a, b)

$$L\|x^n - x^{n-1}\| \|y^{n+1} - y^n\| \leq L\alpha\tau \frac{\|x^n - x^{n-1}\|^2}{2\tau} + \frac{L\sigma}{\alpha\sigma} \frac{\|y^{n+1} - y^n\|^2}{2\sigma},$$

and we choose $\alpha = \sqrt{\sigma/\tau}$, so that $L\alpha\tau = L\sigma/\alpha = \sqrt{\sigma\tau}L < 1$.

Summing the last inequality together with (??) and (??), we get that for any $x \in X$ and $y \in Y$,

$$\begin{aligned} \frac{\|y - y^n\|^2}{2\sigma} + \frac{\|x - x^n\|^2}{2\tau} &\geq [\langle Kx^{n+1}, y \rangle - F^*(y) + G(x^{n+1})] \\ &\quad - [\langle Kx, y^{n+1} \rangle - F^*(y^{n+1}) + G(x)] \\ &\quad + \frac{\|y - y^{n+1}\|^2}{2\sigma} + \frac{\|x - x^{n+1}\|^2}{2\tau} \\ &\quad + (1 - \sqrt{\sigma\tau}L) \frac{\|y^n - y^{n+1}\|^2}{2\sigma} + \frac{\|x^n - x^{n+1}\|^2}{2\tau} \\ &\quad - \sqrt{\sigma\tau}L \frac{\|x^{n-1} - x^n\|^2}{2\tau} \\ &\quad + \langle K(x^{n+1} - x^n), y^{n+1} - y \rangle - \langle K(x^n - x^{n-1}), y^n - y \rangle. \end{aligned} \quad (14)$$

Let us now sum (??) from $n = 0$ to $N - 1$: it follows that for any x and y ,

$$\begin{aligned}
 & \sum_{n=1}^N [\langle Kx^n, y \rangle - F^*(y) + G(x^n)] - [\langle Kx, y^n \rangle - F^*(y^n) + G(x)] \\
 & + \frac{\|y - y^N\|^2}{2\sigma} + \frac{\|x - x^N\|^2}{2\tau} + (1 - \sqrt{\sigma\tau}L) \sum_{n=1}^N \frac{\|y^n - y^{n-1}\|^2}{2\sigma} \\
 & + (1 - \sqrt{\sigma\tau}L) \sum_{n=1}^{N-1} \frac{\|x^n - x^{n-1}\|^2}{2\tau} + \frac{\|x^N - x^{N-1}\|^2}{2\tau} \\
 & \leq \frac{\|y - y^0\|^2}{2\sigma} + \frac{\|x - x^0\|^2}{2\tau} + \langle K(x^N - x^{N-1}), y^N - y \rangle,
 \end{aligned} \tag{15}$$

where we have used $x^{-1} = x^0$. Now as before, $\langle K(x^N - x^{N-1}), y^N - y \rangle \leq \frac{\|x^N - x^{N-1}\|^2}{2\tau} + (\tau\sigma L^2) \frac{\|y - y^N\|^2}{2\sigma}$, and it follows

$$\begin{aligned}
 & \sum_{n=1}^N [\langle Kx^n, y \rangle - F^*(y) + G(x^n)] - [\langle Kx, y^n \rangle - F^*(y^n) + G(x)] \\
 & + (1 - \sigma\tau L^2) \frac{\|y - y^N\|^2}{2\sigma} + \frac{\|x - x^N\|^2}{2\tau} + (1 - \sqrt{\sigma\tau}L) \sum_{n=1}^N \frac{\|y^n - y^{n-1}\|^2}{2\sigma} \\
 & + (1 - \sqrt{\sigma\tau}L) \sum_{n=1}^{N-1} \frac{\|x^n - x^{n-1}\|^2}{2\tau} \leq \frac{\|y - y^0\|^2}{2\sigma} + \frac{\|x - x^0\|^2}{2\tau}.
 \end{aligned} \tag{15}$$

First we choose $(x, y) = (\hat{x}, \hat{y})$ a saddle-point in (15). Then, it follows from (2) that the first summation in (15) is non-negative, and point (a) in Theorem 1 follows. $\square$

8 Experimental Results

References

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